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Narula Institute of Technology
An Autonomous Institute under MAKAUT
2024
END SEMESTER EXAMINATION - ODD 2024
M(CS)101 - ENGINEERING MATHEMATICS - I

TIME ALLOTTED: 3Hours**FULL MARKS: 70**

Instructions to the candidate:

Figures to the right indicate full marks.

Draw neat sketches and diagram wherever is necessary.

Candidates are required to give their answers in their own words as far as practicable

Group A
(Multiple Choice Type Questions)

Answer any ten from the following, choosing the correct alternative of each question: 10×1=10

1.i) If $\phi(x, y) = 0$, then $\frac{dy}{dx} =$

(1) CO1 BL1

a) $\frac{\phi_x}{\phi_y}$

b) $\frac{\phi_y}{\phi_x}$

c) ϕ_y

d) ϕ_x

1.ii) The degree of the homogeneous function $f(x, y) = \frac{\sqrt[3]{x} + \sqrt[3]{y}}{\sqrt{x} - \sqrt{y}}$

(1) CO1 BL2

a) $\frac{1}{3}$

b) $\frac{-1}{2}$

c) $\frac{1}{6}$

d) $-\frac{1}{6}$

1.iii) The rank of the matrix $A = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}$ is

(1) CO1 BL1

- a) 0
- b) 1
- c) 2
- d) 4

1.iv) If α is an eigen vector of the matrix A (if exists) corresponding to a non-zero eigen value λ , then the eigen vector and eigen value of A^{-1} is respectively

(1) CO1 BL2

- a) $\frac{1}{\alpha}$ and $\frac{1}{\lambda}$
- b) α and $\frac{1}{\lambda}$
- c) $\frac{1}{\alpha}$ and λ
- d) α and λ

1.v) For the system of equations $x + 2y + 3z = 1; 2x + y + 3z = 2; 3x + 6y + 9z = 3$ has

(1) CO3 BL3

- a) Unique solution
- b) many solution
- c) No solution
- d) none

1.vi) The Maclaurin's expansion of $\sin x$ is $\sin x = x - \frac{x^3}{3!} - \frac{x^5}{K} - \dots \infty$. The value of K is

(1) CO1 BL1

- a) 4!
- b) 5!
- c) -4!
- d) -5!

1.vii) If $f_x(a, b) = f_y(a, b) = 0$, then (a, b) point is

(1) CO1 BL1

- a) saddle point
- b) point of extreme
- c) critical point
- d) isolated point

1.viii) The value of $\int_0^{\pi/2} \int_0^2 r dr d\theta$

(1) CO3 BL3

- a) $\frac{\pi}{2}$
- b) 2
- c) π

d) 4π 1.ix) The eigen value of $A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$

(1) CO1 BL2

- a) 2
 b) 1,2
 c) 2,3
 d) 1,3

1.x) $\int_2^3 \int_4^6 dx dy =$

(1) CO2 BL2

- a) 1
 b) 2
 c) 12
 d) NONE

1.xi) If $\lambda^3 - 6\lambda^2 + 9\lambda = 4$ is the characteristic equation of a square matrix A, then A^{-1} is

(1) CO2 BL2

- a) $A^2 - 6A + 9I$
 b) $\frac{1}{4}A^2 - \frac{3}{2}A + \frac{9}{4}I$
 c) $\frac{1}{4}A^2 - \frac{3}{2}A + \frac{9}{4}$
 d) $A^2 - 6A + 9$

1.xii) If a function $f(x, y)$ has minimum or maximum value at the points $(2, 3)$ then $f_x(2, 3) =$

(1) CO1 BL1

- a) 0
 b) 1
 c) -1
 d) None

Group B
(Short Answer Type Questions)

(Answer any three of the following) 3x5=15

2. Find the rank of $A = \begin{pmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{pmatrix}$

(5) CO1 BL1

3. Given the function $f(x, y) = \frac{xy(x^2 - y^2)}{x^2 + y^2}, (x, y) \neq (0, 0)$
 $= 0, (x, y) = (0, 0)$

(5) CO2 BL2

Find, from definition, $f_{xy}(0, 0)$ and $f_{yx}(0, 0)$. Verify whether $f_{xy}(0, 0) = f_{yx}(0, 0)$.

4. Evaluate $\int_C (x^2 + y^2) dx - 2xy dy$, where curve C is the rectangle on the XY- plane bounded by $x=0, x=a, y=0, y=b$ (5) CO2 BL3
5. Use Mean value Theorem to prove the following inequalities (5) CO4 BL4
- $$\frac{x}{1+x^2} < \tan^{-1}x < x, \quad \text{for } 0 < x < \frac{\pi}{2}$$
6. Evaluate $\iint_R (x^2 + y^2) dx dy$, where R be the region bounded by $y = x^2, x = 2$ and $y = 1$ (5) CO3 BL3

Group C
(Long Answer Type Questions)
(Answer any three of the following) 3x15=45

- 7a. 1. Investigate for what values of λ and μ the system of simultaneous equations (5) CO4 BL4
- $$\begin{aligned} x + y + z &= 6 \\ x + 2y + 3z &= 10 \\ x + 2y + \lambda z &= \mu \end{aligned}$$
- have (i) No solution (ii) A unique solution (iii) an infinite number of solutions.
- 7b. Find whether the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$ exist. Hence find whether the function is continuous at (0,0). (5) CO2 BL1
- 7c. If $u = xf\left(\frac{y}{x}\right) + g\left(\frac{y}{x}\right)$, then show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = xf\left(\frac{y}{x}\right)$ and $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = 0$. (5) CO2 BL2
- 8a. Expand e^x in power series $(x-1)$ (5) CO2 BL2
- 8b. Examine for consistency and solve, if consistent (5) CO2 BL1
- $$\begin{aligned} 2x - 3y + 5z &= 1 \\ 3x + y - z &= 2 \\ x + 4y - 6z &= 1 \end{aligned}$$
- 8c. Evaluate, by using the changing the order of integration: $\int_0^2 dy \int_y^2 e^{x^2} dx$ (5) CO1 BL1
- 9a. If $\theta = t^n e^{\frac{-r^2}{4t}}$, find what value of n will make $\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \theta}{\partial r} \right) = \frac{\partial \theta}{\partial t}$. (5) CO3 BL2
- 9b. Find the extrema of the function $x^3 + y^3 - 63(x+y) + 12xy$. (5) CO2 BL3
- 9c. If $u = \tan^{-1} \frac{x^2 + y^2}{x+y}$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \frac{1}{2} \sin 2u$ (5) CO4 BL2

- 10a. Use Mean value theorem to prove $0 < \frac{1}{x} \log \frac{e^x - 1}{x} < 1$. (5) CO3 BL3
- 10b. If $u = f(y - z, z - x, x - y)$, prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$ (5) CO3 BL2
- 10c. If $f(x,y) = \frac{x+y}{1-xy}$ and $g(x,y) = \tan^{-1}x + \tan^{-1}y$, find the Jacobean $\frac{\partial(f,g)}{\partial(x,y)}$. (5) CO2 BL2
- 11a. Use mean value theorem to prove that $\frac{x}{1+x^2} < \tan^{-1}x < x$ when $0 < x < \frac{\pi}{2}$ (7) CO3 BL3
- 11b. Evaluate: $\int_0^2 \int_y^2 e^{x^2} dx dy$ (8) CO3 BL3

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